

Scientific Reviewer

Solid CoRL'17 · method aged better than HSR stack

Label recoveries by disturbing the expert, not by deploying the learner.

49/79/72
HSR

3×
Humanoid

−5% / −80%
collect

+62%
rel

Lemma+Prop
theory spine

Claim under review

What the paper asserts and what we stress-test

PRIMARY CLAIM

Optimized supervisor-side action noise yields recovery demos that close covariate shift $\approx$ DAgger robustness with safer/cheaper collection, and lifts human BC on HSR clutter ($\sim +62\%$ rel).

Theory

Lemma 4.1: $\text{shift} \leq T\sqrt{(\frac{1}{2} \text{KL})}$
Prop 4.1: Gaussian noise strictly
 $\downarrow$ KL vs deterministic BC when
learner error > 0

Sim

MuJoCo $\times 4$: DART $\approx$ DAgger
reward; $3\times$ wall-clock Humanoid;
collect -5% vs -80% ; isotropic fails

Real

4 humans HSR:
BC 49 · DART-3 79 · DART-6 72
 α inverted-U honest

Strengths

What earns CoRL acceptance and syllabus retention

Clean positioning

Off-policy alternative to BC vs DAgger with crisp problem split (Shift vs Loss).

Theory spine

Lemma 4.1 + Prop 4.1 give a usable KL story for why *some* Gaussian action noise beats pure BC.

Dual supervisors

Algorithmic TRPO *and* human teleop — not sim-only storytelling.

Noise optimization matters

Isotropic + Wishart ablations show Σ shape/scale are load-bearing.

Honest $\alpha=6$

Larger noise not always better for humans — inverted-U reported, not hidden.

Ops metrics

Wall-clock and collect-time reward reported alongside normalized return.

Weaknesses (1) — design & stats

Punish lightly; demand fixes

n=4 humans · 20 evals

HSR suggestive, not definitive. Demand more subjects / CIs / power analysis.

Bounded- l theory vs L2 practice

Lemma assumes $l \in [0,1]$; practice uses squared L2 — normalize story underspecified.

Half-Cheetah $|X|=117$

Atypical vs modern Gym ~ 17 — reproducibility caveat (custom obs or typo).

Curves without seed CIs

Main-text MuJoCo plots normalized; tabulated multi-seed stats thin.

Weaknesses (2) — fairness & assumptions

Comparisons and theory-practice gaps

DAgger-B fairness

Update-frequency $\times$ algorithm confound; $\beta=0.5$ and schedules always debatable.

Supervisor-unaffected assumption

Paper notes humans may violate; theory still leans on it for KL claims.

No physical DAgger

Human study never runs on-policy DAgger — fairness gap for “safer than DAgger” on robots.

Prop 4.1 sharpness

“KL infinite for deterministic BC” is a support argument — informative but sharp.

Evidence board — Fig. 2 + Fig. 3

Hold claims to plots

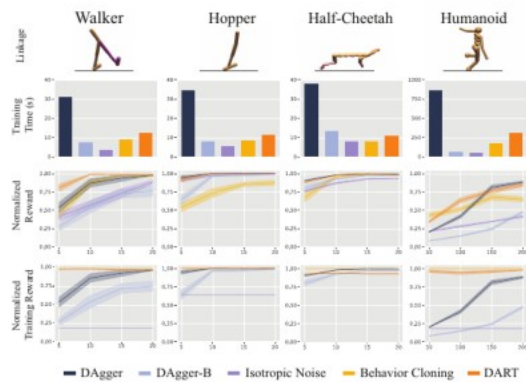


Figure 2: Top: The four different locomotive domains in MuJoCo we evaluated DART on: Walker, Hopper, Half-Cheetah and Humanoid. Top Middle: The time, in seconds, to achieve the performance level reported below. DART achieves similar performance to DAgger in all 4 domains, but requires significantly less computation because it doesn't require retraining the current robot policy after each demonstration. DAgger-B reduces the computation required by less frequently training the robot, but suffers significantly in performance in domains like Humanoid. Bottom Middle: The learning curve with respect to reward obtained during training with each algorithm plotted across the number of demonstrations. Bottom: The reward obtained during collection of demonstrations for learning. DART receives near the supervisor's reward at all iterations whereas DAgger can be substantially worse in the beginning.

policies in Tensorflow. At each iteration we collected one demonstration. In order to make the task challenging for Imitation Learning, we used the same technique as in [6], which is to sub-sample 50 state and control pairs from the demonstrated trajectories, making the learners receive less data at each iteration.

We compare the following five techniques: Behavior Cloning, where data is collected from the supervisor's distribution without noise injected; DAgger, which is an on-policy method that stochastically samples from the robot's distribution and then fully retrains the robot's policy after every demonstration is collected; DAgger-B, which is DAgger but the retraining is done intermittently to reduce computation time; Isotropic-Gaussian which injects non-optimized isotropic noise; and DART. See supplemental material for how the hyper-parameters for each method are set.

We used the following MuJoCo environments: Walker, Hopper, Humanoid, and Half-Cheetah. The size of the state and control space for each task is $\{|X| = 17, |U| = 6\}$, $\{|X| = 11, |U| = 3\}$, $\{|X| = 376, |U| = 17\}$, $\{|X| = 117, |U| = 6\}$, respectively. All experiments were run on a MacBook Pro with a 2.5 GHz Intel Core i7 CPU. We measured the cumulative reward of each learned policy by rolling it for 500 timesteps, the total computation time for each method and the cumulative reward obtained during learning.

Fig. 2 shows the results. In all domains, DART achieves parity with DAgger, whereas Behavior Cloning and DAgger-B are below this performance level in Walker and Humanoid. For Humanoid, DART is 3x faster in computation time and during training only decreases the supervisor's cumulative reward by 5%, whereas DAgger executes policies that have over 80% less cumulative reward than the supervisor. The reason for this is that DAgger requires constantly updating the current robot policy and forces the robot into sub-optimal states. While, one way to reduce this computation is to decrease



Figure 3: Left: Experimental setup for the grasping in clutter task. A Toyota HSR robot uses a head-mounted RGBD camera and its arm to push obstacle objects out of the way to reach the goal object, a mustard bottle. The robot's policy for pushing objects away uses a CNN trained on images taken from the robot's Primesense camera, an example image from the robot's view point is shown in the orange box. Right: The Success Rate for Behavior Cloning, DART($\alpha = 3$) and DART($\alpha = 6$). DART($\alpha = 3$) achieves the largest success rate.

the number of times the policy is updated. DAgger-B illustrates that this can significantly deteriorate performance. Lastly, naively applying isotropic noise does not perform well, and leads to unsafe policies during execution, which suggests the need for optimizing the level of noise.

5.2 Robotic Grasping in Clutter

We evaluate DART with human supervisors in a grasping in clutter task on a Toyota HSR robot. We consider a task inspired by the Amazon Picking Challenge [2]. The goal of the task is for the robot to retrieve a goal object from a cupboard. The task is challenging because the goal object is occluded by obstacle objects and the robot must reason about how to clear a path based on observations taken from an eye-in-hand perspective. The objects are 6 common household food items, which consist of boxes and bottles with varying textures and mass distributions. The target object is fixed to always be a mustard bottle. The robot, task and image viewpoint are shown in Fig. 3. See supplemental material for additional information on the task.

We use 4 supervisors who have robotics experience but not specifically in the field of Imitation Learning and compare Behavior Cloning and DART. When performing the study, we first collect $N = 10$ demonstrations with Behavior Cloning (i.e. no noise) and then in a counter-balanced ordering collect $N = 30$ more demonstrations with each technique. Our experiment was within-subject, have every supervisor perform all three methods.

We only updated the noise parameter after the first 10 demonstrations (i.e. $K = 2$). The final robot policy was then trained on the total of 40 demonstrations. We consider two different choices of α : $\alpha = 3\text{Tr}(\Sigma_1)$ and $\alpha = 3\text{Tr}(\Sigma_2)$. These choices correspond to the intuition that with small datasets in high dimensional image space the robot will have significant shift from the current loss on the supervisor's distribution. Through the rest of the paper, we will write our choices as $\alpha = 3$ and $\alpha = 6$ for brevity in notation.

During policy evaluation, we measured success as 1) the robot is able to identify the goal object and 2) there being a clear path between its gripper and the object. Once these conditions are identified, as detailed in the supplemental material, an open-loop motion plan is generated to execute a grasp around the target object.

In Fig. 3, we report the average performance of the three techniques across DART 20 times on different sampled initial states. DART with $\alpha = 3$ performs the best out of the three techniques, with a 79% success rate over traditional Behavior Cloning's 49% success rate. Interestingly, DART with $\alpha = 6$ performs better than Behavior Cloning, but only has a 72% success rate. This may suggest this level of noise was potentially too high for the human supervisor.

6 Conclusion

This paper considers injecting noise into the supervisor's demonstrations to minimize covariate shift. Our algorithm, DART, provides the robot with corrective examples at the boundary of the supervisor's policy. By being concentrated around the supervisor's policy, it collects demonstrations without visiting highly sub-optimal states and is easier for human supervisors tele-operating. We demonstrate it achieves parity with on-policy methods in simulated MuJoCo domains and is significantly better than traditional Behavior Cloning on a real robot. For code and more information see <http://berkeleyautomation.github.io/DART/>

Theory stress-test

What holds; what to tighten

Lemma 4.1

If $I \in [0, 1]$, covariate shift $\leq T \sqrt{(1/2 D_{\text{KL}}(\text{robot traj} \parallel \text{noisy-supervisor traj}))}$. Minimizing KL $\Rightarrow$ maximizing likelihood of learner actions under noisy expert.

Prop 4.1: with Gaussian action noise, noisy expert is strictly closer in KL than pure BC whenever learner error > 0 .

Keep

KL objective $\rightarrow$ closed-form $\tilde{\Sigma}$ is elegant
and teaches why *shape* of noise matters.

Tighten

Bridge bounded- $I \leftrightarrow L2$ explicitly;
state Half-Cheetah obs; multi-seed CIs;
physical DAgger baseline or justify gap.

Verdict

Accept · syllabus cornerstone · punish overclaim on HSR generality

VERDICT

Solid CoRL'17 paper. Methodological contribution (noise-in-demo for shift) aged better than the specific HSR stack. Suitable Data syllabus cornerstone for recovery augmentation.

Accept contribution · demand statistical humility on $n=4/20$ · do not market 2017 CNN as 2026 VLA prior.

Aged well

Σ^α recipe · α honesty ·
collect-safety metrics ·
ACT still cites it

Aged poorly

HSR open-loop grasp ·
TF+MacBook timings ·
Half-Cheetah dim quirk

Use as

Recovery-as-data reading
paired with Ross&Bagnell
+ one chunking paper

Actionable fixes list

What a revision / modern re-run should do

1. Tabulate MuJoCo returns $\pm$ CI across ≥ 5 seeds; publish exact wall-seconds.
2. Clarify Half-Cheetah observation dim (117 vs Gym 17) in errata / appendix.
3. Bridge lemma's bounded loss to L2 practice (normalize / cite).
4. Power analysis / more subjects for human study; report per-supervisor.
5. Matched physical DAgger arm *or* explicit "no unsafe human DAgger" justification.
6. Align DAgger-B update schedule ablations factorially with DART K.
7. Modern transfer card: ALOHA/DROID teleop + α sweep (Empiricist P0).
8. Label ACT relationship: complementary to chunking, not obsolete.

Exam bite

One paragraph a reviewer can grade

EXAM BITE

DART is a solid CoRL'17 methodological paper: optimized supervisor-side action noise closes covariate shift with safer/cheaper collection than DAgger and lifts HSR BC 49→79 ($\alpha=3$). Strengths = theory+dual supervisors+isotropic honesty. Weaknesses = $n=4/20$, bounded-I vs L2, Half-Cheetah dim, DAgger-B fairness, supervisor-unaffected assumption. Verdict: accept contribution; method aged better than HSR stack; demand statistical humility.

Tattoo card

HSR 49% / 79% / 72% · +62% rel · Humanoid 3× wall-clock · collect −5% (DART) vs −80% (DAgger) · α inverted-U

49%

BC

79%

$\alpha=3$

72%

$\alpha=6$

+62%

rel

3×

wall

−5% vs −80%

collect reward

$\Sigma^{\alpha} \gg I$

opt > isotropic

α inv-U

3 > 6 humans

Close — Reviewer

Accept + punish overclaim

1. Accept methodological contribution: recovery-as-data via optimized action noise.

2. Praise dual supervisors, isotropic/Wishart ablations, honest $\alpha=6$.

3. Punish thin human n, theory-practice loss mismatch, dim quirk, fairness gaps.

4. Syllabus: Data cornerstone; pair with Ross&Bagnell + ACT chunking.